Sam Belliveau

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Education

Cornell University — Ithaca, NY
B.S. in Electrical & Computer Engineering — *Expected May 2026*

- **GPA:** 3.7 / 4.3

Related Courses: Discrete Structures Computer Science I - III, Digital Logic & Computer Organization, Differential Equations, Linear Algebra, Signals & Systems, Phyisics Mechanics, Data Science for Engineers, Electromagnetic Fields and Waves

Languages: C, C++, C#, Python, Java, Rust, Swift, Go, JavaScript, Verilog HDL, SQL, HTML, GLSL, Bash / ZSH
Technologies: Git, Linux, Docker, VS Code, Robot OS (ROS), Cassandra, BigQuery, Numpy, Scipy, PyTorch

Experience

Cornell University (Prof. Abe Davis Lab) — Ithaca, NY
Undergraduate Researcher — *June 2023 – Present*

- **CaptureGraph (System Architect):** Developed a domain-specific language (DSL) that serializes experimental procedures into a graph data model. Built a companion iOS app that interprets this graph to provide state-aware AR guidance and real-time data validation, eliminating the need for hard-coded capture apps.
- **ReCapture (Modernization):** Refactored 7,000+ lines of legacy Swift code to enable HDR and RAW photography. Identified the "rigidity" bottleneck in bespoke apps which motivated the pivot to the programmable CaptureGraph framework.
- **CineCraft (Mobile Vision):** Addressed a critical HCI/Signal-Processing challenge where subject tracking fought with zoom automation. Implemented a stabilization filter using Apple's Vision framework to enable smooth "virtual focus puller" capabilities.

Robotics Software Engineer — November 2023 – Present
Cornell University Autonomous Underwater Vehicles — *Ithaca, NY*

- Slashed CPU usage by 80% by offloading Kalman Filter computations to the GPU using Nvidia's CUDA framework
- Enhanced autonomous navigation precision by implementing least squares optimization for dynamic thruster allocation

Signal Analysis Intern — January 2023 – May 2023
Feinstein Institute for Medical Research — *Manhasset, NY*

- Engineered an automated Sharp Wave Ripple (SWR) detection pipeline using Python and SciPy to process large-scale EEG datasets
- Conducted analysis on SWR characteristics to refine detection algorithms, directly contributing to improved seizure prediction models

VEX Robotics Coach / Mentor — October 2022 – May 2023
PLAYIDEAs NY. — *Great Neck, NY and Manhattan, NY*

- Coached 3 local VEX robotics of 6-8 students each, teaching them the basics of robotics and programming
- Developed lesson plans to teach students about PID control, Odometry, and very simple digital filtering

President of Software Engineering — December 2018 – June 2022
StuyPulse Robotics — *Manhattan, NY*

- Led and taught team of ~50 members to write software for a 120lb robot that competed in FRC Championships
- Implemented, taught, and documented PID control, Odometry, Digital Filtering, and Motion Profiling
- Communicated software design to competition judges and won 4 Innovation in Control Awards

Projects

UCI Chess Engine in Rust | *Rust, Min Max, Alpha Beta Pruning, Chess* — June 2023 – July 2023

- Developed a high-performance UCI-compliant Chess Engine in Rust, achieving an estimated Elo of 2700 (Grandmaster level)
- Optimized search efficiency using Alpha-Beta Pruning and iterative deepening, significantly boosting engine strength

StuyLib | *Java, JavaDocs, JitPack, Control Theory, Digital Filtering* — January 2020 – August 2022

- Initiated and led the development of StuyLib, an award-winning Control Theory Library
- Heavily maintained and documented the library to ensure its longevity and ease of use